

Agentic-AI driven Cybersecurity Threat Detection

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1 Introduction and Literature Survey

1.1 Background and Context

Artificial Intelligence and Machine Learning have made it possible for automated systems to find patterns in network traffic. Many Artificial Intelligence security systems still need people to help them and they have a hard time keeping up with new threats.

In recent days, Agentic AI is a way to make cybersecurity systems that can work on their own. These Agentic AI systems can look at the environment, make decisions and stop threats all by themselves. They can do this because they have capabilities like reasoning and planning. This means they can do cybersecurity tasks without any help. Some studies show that AI can make threat detection more accurate and help with real-time defense against cyber threats [7], [13].

Agentic AI extends generative AI by improving reasoning, planning, tool use, memory, and autonomy that helps in enabling goal-directed, long-horizon behavior across real environments. In cybersecurity this is really useful for continuously monitoring and responding quickly to threats. At the same time they expand the attack surface through prompt/-tool injection, unsafe action execution, memory poisoning, and multi-agent manipulation. [1][2].

For financial and retirement platforms, it's found that the agentic methods do complement the deep anomaly detection and UEBA. They help protect the flows, where the fraud likelihood is highest, such as rollovers, disbursements, and withdrawals. Increased use of CNN/LSTM/Transformer ensembles paired with event-driven and SOAR, zero-trust as governance agentic layers are seen in Architectures.[3]

1.2 Motivation and Importance of the Study

AI is being adopted increasingly by the threat actors themselves to automate reconnaissance, social engineering and exploitation. Under high volume of attacks and novelty, traditional signature or rule pipelines lag. For the finance domain transparency, safe autonomy and governance are essential for adoption; though proactive, autonomous agents promise anomaly surfacing, reduced analyst load and faster time-to-contain.[3][4]

As indicated from the literature survey autonomous Agentic AI systems improve coverage and response, that use deep ensembles of CNN+LSTM+Transformers, have benefits over traditional threat detection systems where it can; learn spacial-temporal patterns in logs/transactions, create UEBA baselines per user/entity, have streaming risk scoring integrated with SOAR playbooks, based on event-driven architectures.[3] These are narrowly scoped agents that align to NIST CSF 2.0 for better scalability and safety than a monolithic end-to-end agents. Agents reduce the failure impact on live SOCs.[1][2]

Agentic AI systems that can act on their own have benefits over traditional security systems. They can watch what is happening on the network, find activities and take action to stop threats without someone always supervising them. These systems can look at a lot of security data, find patterns that do not seem right and change how they work as new threats appear.

Recent studies have shown that Agentic AI systems that can act on their own can make cybersecurity better by allowing them to think and act independently, continuously monitor and find threats[8], [9]. This capability makes Agentic AI systems a great option for the next generation of cybersecurity solutions.

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1.3 Problem Definition and Scope

Problem: The development of a cybersecurity framework that is capable of automatically detecting cyber threats and responding to them using Agentic AI techniques for high-stakes platforms.

In-scope: (i) defensive agentic workflows (monitoring, detection, triage, response), (ii) adversarial capabilities unique to agents (reasoning compromise, unsafe tool use, memory/state poisoning, multi-agent manipulation), (iii) governance and evaluation (CSF 2.0 adaptation, specialized benchmarks).

Out-of-scope: cryptographic payload design and physical robotics beyond cyber. [1][2][5]

1.4 Research Objectives and Questions

1. Synthesize threats, defenses, and evaluation for agentic AI in cybersecurity. [1][2] Compare detection/response architectures for finance (UEBA, deep ensembles, transformers, federated learning, agentic orchestration).[3] Map controls to NIST CSF 2.0 and OWASP agentic threat model; identify least-agency guardrails. [5]. Assess RL-based response automation for latency and adaptability, and its risks. [6]
2. **Agentic AI Effectiveness:** Analyze how agentic AI systems detect cybersecurity threats by continuous monitoring network activities and identifying anomalous patterns in complex environments. This objective identifies how agent-based frameworks improve detection accuracy compared to traditional machine learning approaches [7], [13].
3. **Autonomous Threat Detection:** Investigate methods that enable autonomous cybersecurity agents to detect threats in real time without human intervention. Agentic AI systems analyze and understand system behavior, reason about threats, and automatically initiate defensive actions, improving cybersecurity responsiveness [9], [11].
4. **Multimodal Data Integration:** Examine how agentic AI models can integrate multiple data sources such as network traffic logs, system logs, threat intelligence feeds, and sensor data to enhance anomaly detection performance. Multimodal analysis can improve the identification of complex and coordinated cyberattack patterns [15].
5. **Adaptive Learning and Decision-Making:** Evaluate how agentic AI systems continuously learn from evolving cybersecurity environments and adapt detection strategies by updating rules and decision policies based on new threat patterns and feedback [14].
6. **Security and Governance Challenges:** Investigate challenges in deploying agentic AI for cybersecurity, including scalability, safety, transparency, ethical governance, and resilience against adversarial manipulation of AI models [8], [9].

1.5 Expected Contributions

1. **Thematic:** The thematic survey is really important because it looks at all the papers with a focus on the finance sector. It has tables that compare techniques and benchmarks and it also looks at governance controls. The survey uses a secure by design reference pattern that combines orchestration and least agency permissions and sandboxed tools and federated learning. This is useful because it helps identify

research gaps and it has a blueprint for evaluation that is based on process metrics [1][2][3][4][5][6].

2. **Agentic AI-Based Threat Detection Framework:** Develop an autonomous cybersecurity architecture using agentic AI to continuously monitor network activities, detect anomalies, and respond to threats through reasoning, planning, and adaptive decision-making without constant human supervision [7], [13].
3. **Autonomous Multi-Agent Cyber Defense Architecture:** Design a multi-agent cybersecurity framework where many intelligent agents work together to detect anomalies and analyze threats and respond to incidents. This is useful because it can be used in cloud and enterprise environments and it can be scaled up to monitor threats.[11], [15].
4. **Adaptive Anomaly Detection Mechanisms:** Develop adaptive anomaly detection methods that can adjust to new cyberattack patterns and learn from the environment. This can be done using AI systems that update their threat models and detection thresholds as they learn. [10], [14].
5. **Comparative Analysis of AI-Driven Cybersecurity Techniques:** We need to look at machine learning and deep learning and agentic AI to understand which one is best at detecting threats accurately, automate the threat detection and responding to them efficiently[9], [13].
6. **Secure and Governable Autonomous Cybersecurity Systems:** Investigate the security and governance and ethical challenges of using AI and propose strategies to improve transparency and reliability and resilience against attacks and system failures. [8], [9].

1.6 Review Methodology and Source Selection Criteria

The literature review for this study was performed through a systematic search of major academic databases and scientific repositories to identify relevant research related to **Agentic Artificial Intelligence, anomaly detection, and AI-driven cybersecurity systems**. The main sources used for identifying relevant publications include IEEE Xplore, ACM Digital Library, ScienceDirect, arXiv, Google Scholar, and other recognized research repositories.

Some keywords used during the literature search included: "agentic AI cybersecurity," "an agentic security taxonomy/defenses/evaluation," "autonomous cybersecurity agents," "AI-driven threat detection," "anomaly detection in cybersecurity," "multi-agent security systems," "LLM-based anomaly detection," "autonomous cyber defense," and "AI-based security monitoring."

The literature review focused on identifying research that explores and helps to learn the application of **Agentic AI systems, anomaly detection techniques, and autonomous cybersecurity architectures** for detecting and responding to cyber threats in modern digital infrastructures.

1.6.1 Inclusion Criteria

The following criteria were checked to select relevant research papers:

1. Publications from 2025-2026 focusing on recent advancements in agentic AI and cybersecurity

2. Peer-reviewed journal articles, conference papers, and recognized research preprints
3. Studies focusing on Agentic AI architectures, anomaly detection systems, and autonomous cybersecurity agents
4. Research addressing cybersecurity threat detection, automated incident response, and adaptive AI-based security systems
5. Works discussing multimodal anomaly detection, multi-agent cybersecurity frameworks, or AI-driven predictive analytics

1.6.2 Exclusion Criteria

The following criteria were checked to exclude irrelevant studies:

1. Publications not related to Agentic AI or cybersecurity applications
2. Research focusing exclusively on theoretical AI models without practical cybersecurity relevance
3. Studies lacking sufficient methodological details or experimental validation
4. Research addressing outdated cybersecurity techniques without modern AI-based approaches

After conducting the literature search, multiple relevant publications were identified and analyzed. From these sources, **some of the research papers were selected for detailed review** based on their direct relevance to Agentic AI-based cybersecurity systems, anomaly detection frameworks, and autonomous cyber defense mechanisms.

Priority was given to studies that provide insights into **autonomous threat detection, multi-agent cybersecurity architectures, and intelligent anomaly detection systems**, as these topics directly support the research objectives of developing an **Agentic AI-driven cybersecurity threat detection framework**.

1.7 Survey of Existing Approaches

1.7.1 Early Solutions and Classical Techniques

There are limitations to the current signature/rule pipelines and classical ML with manual interventions. They offer transparency and speed but struggle with zero-day drift and tactics[3]

1.7.2 Modern Approaches and Evolving Trends

Its indicated from the literature survey autonomous Agentic AI systems improve coverage and response, that use deep ensembles of CNN+LSTM+Transformers,[3] have benefits over traditional threat detection systems where it can; learn spacial-temporal patterns in logs/transactions,create UEBA baselines per user/entity, have streaming risk scoring integrated with SOAR playbooks, based on event-driven architectures.[3] These are narrowly scoped agents that align to NIST CSF 2.0 for better scalability and safety than a monolithic end-to-end agents. Agents reduce the failure impact on live SOCs.[1][2]

Autonomous Agentic AI Systems: Autonomous Agentic AI Systems are like agents that can think, plan and act on their own to defend against cyber threats. They watch what is happening in the system all the time, look for patterns that might be a threat

and respond to those threats in time without constant human supervision [7], [13]. **Multi-Agent Cybersecurity Architectures:** Modern frameworks adopt multi-agent systems where we use smart agents that work together to find anomalies, look at threats and respond automatically. This way we can monitor cloud and enterprise systems and make them safer.[11]

Sequence-Aware Behavioral Detection: Sequence-based anomaly detection models understand the behavioral patterns in the system to find events that might be a cyber-attack in real time [12].

Multimodal Anomaly Detection Systems: Multimodal approaches combine data from multiple sources such as network logs, system events, and sensor data to find complex patterns and improve detection accuracy [15].

Predictive Analytics and Adaptive Learning: Modern cybersecurity systems integrate **predictive analytics and adaptive learning** to learn historical attack patterns and predict emerging threats and enable proactive cyber defense [14].

Overall, cybersecurity research is moving toward **autonomous and adaptive security architectures powered by agentic AI**, combining deep learning, anomaly detection, and multi-agent collaboration for real-time threat detection and response [7], [13].

1.7.3 Emerging Models and Hybrid Frameworks

Federated learning enables cross-institution fraud defense without sharing raw data. This is really helpful because it adds a level of explainability and control. It also makes sure that the system is fair and trustworthy. Federated learning is used with governance(AIGA/SAGA) , which is like a set of rules that makes sure the system is working correctly. This is important for things like finance, where people need to be able to trust the system.[3][6]

Federated learning is also used with something called reinforcement learning which's a type of artificial intelligence. This type of learning helps the system respond quickly to changes. It also raises some concerns about safety and stability. To make sure the system is working correctly people use something called vectorized feature stores, which's like a big database that stores all the information. They also use something called agent-to-agent coordination, which's like a way for different parts of the system to talk to each other.[3][5]

One of the challenges of using reinforcement learning in real-world situations is that it needs to be controlled. This is because the system is making decisions on its own and people need to make sure it is not doing anything wrong. To control the system people use something called guardrails, which's like a set of rules that the system has to follow. They also use something called sim-to-real certification, which's like a test to make sure the system is working correctly in the real world.[6]

Agentic AI-Driven Hybrid Detection Models:Modern frameworks integrate traditional anomaly detection algorithms with **agentic AI systems** to detect and respond to cyber threats. These architectures enable intelligent agents to analyze system behavior and automatically respond to cyber threats, improving real-time security monitoring [7], [13].

Multi-Agent Cybersecurity Frameworks:Systems use **multiple intelligent agents** that work together to detect anomalies and respond to threats. These distributed architectures improve scalability and support monitoring of complex cloud and enterprise environments [11].

Sequence-Aware and Behavioral Anomaly Detection:Hybrid models combine anomaly detection algorithms with **sequence-aware analysis** to detect abnormal behavioral patterns and suspicious system interactions in real time [12].

Multimodal Anomaly Detection Architectures:Multimodal frameworks integrate multiple data sources such as **network traffic, system logs, and sensor data** to detect complex cyberattack patterns that may not be identifiable using a single data source [15].

Predictive and Adaptive Security Models:Hybrid approaches also combine **predictive analytics and adaptive learning**, enabling systems to identify cyber threats and dynamically update detection strategies based on evolving attack patterns [14].

Overall, these hybrid AI architectures combine **anomaly detection, agent-based intelligence, and multimodal data analysis** to create more robust and adaptive cybersecurity systems capable of addressing modern cyber threats [7], [13].

1.8 Comparative Evaluation of Techniques

Table 1. Evolving trends in Cyber security Threat analysis for Finance Use Cases				
Approach	Strengths	Limitations	Finance Use-Cases	Key Sources
Rules/Signatures	Fast, transparent	Miss novel tactics, drift-sensitive	Known IOCs/rollovers policy checks	[3]
Classical ML	Efficient, interpretable	Manual features; limited non-linear patterns	First-pass fraud filters	[3]
Deep Ensembles	High PR-AUC; long-range context	Explainability; compute; drift	Streaming fraud/UEBA scoring	[3]
Agentic Orchestration	Correlates signals; machine-speed action	Autonomy risks; governance needed	SOC triage, playbooks with human-on-the-loop	[1][2]
Federated Learning	Cross-org generalization; privacy	Coordination and policy complexity	Consortium fraud analytics	[3]
RL Response	Latency-optimized, adaptive	Stability; sim-to-real; adversarial learning	Automated containment actions	[6]

Table 2. Comparative analysis on various parameters				
Stack Element	Governance Control	Injection Resilience	Explainability	Ops Cost
Rules/Signatures [3]	Policy-as-code; change control	Low (content spoofing)	High	Low
Deep Ensembles [3]	Model registry; drift monitors	Medium with sanitization	Medium (XAI needed)	Medium
Agentic Orchestration [1][2][5]	Least-agency; immutable logs	High with sandboxing & intent binding	Medium (trace replay)	Medium-High
RL Response [6]	Action allow-lists; graded autonomy	Medium (adversarial training required)	Low-Medium	High (training/ranges)

Table 3. Comparative analysis of Agentic AI-based techniques for cybersecurity threat detection

Method / Technique	Architecture	Application Area	Data Source	Key Outcome
Agentic AI Cybersecurity Automation	Autonomous AI agents	Cyber threat monitoring	Network logs, threat intelligence	Enables automated threat detection and response
Agentic AI System Framework	Autonomous AI architecture	Intelligent decision systems	System activity data	Adaptive AI systems capable of reasoning and planning
Cognitive Autonomous Cyber Defense	Agentic AI security model	Security governance	Security event logs	Autonomous cyber defense and threat monitoring
Autonomous Anomaly Management	Agent-based anomaly detection	Complex system monitoring	System behavior data	Automatic anomaly identification in dynamic environments
Adaptive Cybersecurity Architecture	Multi-agent security framework	Digital ecosystems	Network traffic data	Collaborative agent-based threat detection
Sequence-Aware Anomaly Detection	Sequence-based AI model	Autonomous system monitoring	Activity sequences	Detection of abnormal AI-agent behaviors
Multi-Agent Cybersecurity Framework	Distributed agent architecture	Security operations centers	Network and system logs	Autonomous security monitoring and response
Predictive Anomaly Detection System	Agentic predictive analytics	Data-driven monitoring	Historical and sensor data	Predictive detection of anomalies
Multimodal Agentic Anomaly Detection	LLM-based agent architecture	Multimodal anomaly detection	Logs, sensor data	Improved detection using multiple data modalities

Key observations:

- 1. Autonomous vs Traditional Security Systems:** Traditional intrusion detection systems rely heavily on predefined rules and supervised machine learning models. In contrast, **Agentic AI systems introduce autonomous agents** that can think and plan and respond to threats quickly which helps with real-time cyber defense [7], [13].
- 2. Architecture Variations:** Agentic cybersecurity frameworks adopt different architectures including **single-agent systems, multi-agent frameworks, and multimodal AI architectures**. Multi-agent systems provide improved scalability and enable distributed monitoring of large digital infrastructures [11].
- 3. Data Source Diversity:** Modern cybersecurity systems increasingly rely on **multiple data sources such as network logs, system events, behavioral patterns, and sensor data** to improve anomaly detection accuracy. Multimodal approaches enable detection of complex attack patterns that cannot be identified using a single data source [15].
- 4. Adaptive Learning and Prediction:** Agentic AI models incorporate **adaptive learning and predictive analytics**, allowing cybersecurity systems to continuously learn from evolving attack patterns and anticipate future threats [14].
- 5. Real-Time Cyber Defense:** Agentic architectures enable **real-time detection and automated incident response**, significantly reducing reliance on manual in-

tervention in cybersecurity operations and improving overall system resilience [9], [12].

1.9 Dataset and Benchmark Overview

Table 4. Datasets and Benchmarks used for evaluating AI-driven cybersecurity systems

Dataset/Benchmark	Focus	Metric/Signal	Notes	Source
Credit-Card/Bank-Fraud	Imbalanced fraud detection	PR-AUC, Recall at low FP	Often synthetic; use SMOTE/AE	[3]
ToolEmu	Tool-use safety	Side-effect rate; policy violations	LLM-emulated tools for scale	[2]
OS-Harm	Desktop agents	Safety + accuracy	LLM judges validated vs humans	[2]
AgentDojo	Prompt-injection eval	Attack success rate (ASR)	Dynamic agent tasks	[5]
INJECAGENT	Indirect injections	End-to-end ASR; benign perf	Tool-integrated pipelines	[5]
AgentHarm	Harmful tool use	Compliance & competence	Tool-sequence safety	[5]
R-Judge	Risk awareness	Trajectory risk detection	Judge/guardrail development	[5]
BADROBOT	Embodied agent misuse	Physical tool abuse	Cyber-physical scope	[5]
ARCeR/Cyberwheel	Ranges & sim-to-real	Task success, fidelity	Structured red-blue evaluation	[1][2]

"AgentDojo and Injecagent" benchmarks focus on injections from untrusted external content such as HTML/Accessibility trees and retrieved snippets.[2]. It is essential to report both the success rate of attacks and the benign performance, when evaluating these benchmarks, which is the utility when there is no attack. ,reason being over-defense can limit the capability of the agent [5].The benchmarks OS-Harm and ToolEmu, which complement the webcentric tests, test desktop and tool-emulation behaviors.Both OS-Harm and ToolEmu support repeatable traces which are suitable for judges, like R-Judge to score risk awareness and policy compliance at the trajectory level.[2][5]

Limitations of Existing Datasets:

Despite the availability of benchmark datasets, several challenges remain in cybersecurity dataset research. Many datasets contain **outdated attack patterns**, lack diversity in network configurations, and suffer from **class imbalance**, where normal traffic significantly outnumbers malicious activity. Additionally, most datasets focus on specific attack scenarios and may not fully represent the complexity of real-world cybersecurity environments.

Agentic AI-based systems aim to overcome these limitations by enabling **adaptive learning and real-time anomaly detection**, allowing cybersecurity systems to continuously learn from dynamic environments rather than relying solely on static datasets [7], [13].

1.10 Tools, Frameworks, and Experimental Protocols

Table 5. Framework layers for the Agentic AI Threat Modeling

Layer	Examples	Purpose	Principal Risks	Sources
Modeling	CNN/LSTM/Transformer/CNN/graph	Security anomaly detection	Opacity; drift	[3]
Orchestration	SIEM/SOAR/Event bus	Correlation and automated playbooks	Runaway actions; policy gaps	[3][1]
Governance	AIGA/SAGA/Zero-Trust/OPA	Explainability, audit, policy control	Overhead; alignment gaps	[3][1][5]
Protocols	MCP/A2A	Agent-tool and agent-agent communications	Injection; spoofing; recursive DoS	[2][5]
Isolation	Sandboxing/containers	Constrain tool execution	Overhead; escape risk	[2][5]

Preprocessing Tools:

- **Log Parsing and Feature Extraction Tools:**Security logs, network traffic, and system activity data are processed using log-parsing and feature extraction tools to convert raw cybersecurity data into structured datasets for anomaly detection models. As a result this efficiently analyzes network events and system behavior patterns. [15].
- **Data Cleaning and Normalization Tools:**Data Preprocessing pipelines remove noise, normalize features, and eliminate redundant records to improve the quality of training data used in anomaly detection systems [14].
- **Data Aggregation and Monitoring Systems:**Security monitoring systems collect and aggregate logs from multiple sources such as network devices, servers, and applications to create comprehensive datasets for AI-based cybersecurity models [9].

AI and Machine Learning Frameworks:

- **Python-Based AI Ecosystem:**Python is widely used for implementing cybersecurity threat detection systems due to its extensive ecosystem of machine learning and data analysis libraries.
- **TensorFlow and PyTorch:**These deep learning frameworks support the development of neural network models for anomaly detection and provide GPU acceleration for large-scale data processing.
- **Multi-Agent AI Frameworks:**Agentic AI architectures often use multi-agent systems where autonomous agents collaborate to monitor networks, detect anomalies, and perform automated threat responses [11], [13].
- **Large Language Models (LLMs):**LLM-based agents analyze security logs and threat intelligence data to enhance anomaly detection and enable intelligent decision-making in cybersecurity systems [15].

Training Protocols:

- **Dataset Preparation and Feature Engineering:**Cybersecurity datasets are transformed into structured formats through preprocessing and feature extraction techniques.

- **Training and Testing Splits:**Datasets are typically divided into training, validation, and testing subsets to evaluate the generalization capability of cybersecurity detection models.
- **Hyperparameter Optimization:**Hyperparameters such as learning rate, batch size, and model configurations are optimized using methods like grid search and random search.
- **Cross-Validation:**Cross-validation techniques are applied to improve model reliability and reduce overfitting.

Computational Infrastructure:

- **GPU-Enabled Training Systems:**Deep learning models require GPU-based systems for efficient training and faster computation.
- **Distributed Computing Platforms:**Large-scale cybersecurity datasets may require distributed data processing frameworks for efficient analysis and model training.
- **Cloud Computing Platforms:**Cloud-based machine learning environments provide scalable infrastructure for training and deploying cybersecurity detection models.

Reproducibility Practices:

- **Open Research Datasets and Frameworks:**Public datasets and open-source implementations support reproducible research.
- **Transparent Experimental Documentation:**Detailed documentation of preprocessing pipelines, model architectures, and training procedures ensures transparency in experimental design.
- **Parameter and Configuration Reporting:**Publication of hyperparameter configurations and experimental settings allows other researchers to replicate and validate proposed cybersecurity detection models [14].

1.11 Evaluation Metrics Used in Literature

- **Detection efficacy:** Precision/Recall/F1, PRÁAUC; latency/throughput; calibration; alert reduction. [3]
- **Safety/process:** Completion-under-Policy (CuP), Risk Ratio, refusal rate, attack success rate under indirect injections, judge-human agreement. [2][5]
- **Reliability:** The system passes with accuracy across multiple trials.We use trajectoryÁlevel scoring with LLM/agent judges, sandbox fidelity. [1][2][5]

1.12 Limitations of Prior Work

- **Dataset Limitations:** There are issues with current datasets.They are often imbalanced, noisy and have limited labeled data.Also sharing data across institutions in finance is hard[3]. Many studies use benchmark datasets like KDD Cup 1999 NSL-KDD, CIC-IDS2017 and UNSW-NB15.These datasets may not reflect modern cyber threats and attack patterns[14]. Existing datasets rarely include multimodal security data, such as combined network traffic, system logs, behavioral activity records, and sensor data, limiting the effectiveness of advanced anomaly detection systems [15].

- **Scalability Challenges:** Agentic AI systems often require significant computational resources to process large volumes of security data. Handling massive traffic generated in cloud and distributed infrastructures remains challenging for many anomaly detection models [11]. Communication and coordination overhead between multiple autonomous agents can impact system performance.
- **Generalization Issues:** Models trained on specific datasets often fail to generalize effectively across different network environments. Many detection systems struggle to detect zero-day attacks and novel cyber threats [9]. Limited mechanisms exist for continuous learning and adaptation to evolving attack patterns.
- **Models/agents:** There are gaps in understanding how models make decisions. Models can be affected by changes in data over time. They can also be manipulated by prompts or tools. There exist a multi-agent collusion risks. [1][2][5]. Many AI-based cybersecurity systems function as black-box models, making it difficult to interpret how threat detection decisions are made. Limited explainability reduces trust in automated security systems used in critical environments [13].
- **Operations:** Legacy integration, talent gaps, and transparency/trust barriers for autonomy.[4]. Response time, system integration, data privacy, and regulatory compliance are often insufficiently addressed in existing studies [7]. A gap between research models and real-world cybersecurity deployment environments.
- **Evaluation:** Over-reliance on end-state metrics; limited long-horizon and multi-agent stress testing; sim-to-real gaps for RL. [1][2][6]
- **Security and Adversarial Risks:** AI-based cybersecurity models are vulnerable to adversarial attacks, where malicious inputs are crafted to evade detection. Autonomous AI agents may introduce additional security risks if attackers exploit weaknesses in the agent decision-making process [8].

1.13 Research Gaps and Open Questions

Analysis of the reviewed literature reveals several **research gaps in Agentic AI-driven cybersecurity systems** that require further investigation.

Research Gaps:

- **G1-Least-Agency Execution:** How to bind intent, permissions, and execution context across heterogeneous tools and protocols so injected instructions cannot cascade, without crippling utility? [2][5]
- **G2-Assurance for Long-Horizon, Multi-Agent Systems:** Process-aware metrics, repeated-trial safety, and ranges with realistic agent communication and oversight. [1][2][5]
- **G3-Finance-Specific Adaptation:** Retirement/wealth flows, older-user fraud patterns, and legacy cores via privacy-preserving federation and explainable decisions. [3][4]
- **G4-RL Safety Sim-to-Real**
- **G5-Autonomous Threat Detection:** Most cybersecurity systems still require human supervision for decision-making and incident response. Fully autonomous agent-based cybersecurity frameworks are still limited. [7], [13]

- G6-Multimodal Security Data Integration: Many detection systems rely on single data sources such as network traffic or logs, limiting their ability to detect complex attacks. [15]
- G7-Scalability of Agentic AI Systems: Deploying agent-based cybersecurity systems in large cloud and distributed infrastructures introduces scalability and coordination challenges. [11]
- Gap 4: Adversarial Robustness: AI-based cybersecurity models can be vulnerable to adversarial attacks designed to evade detection systems. [8]
- Gap 5: Continuous Learning: Many detection models struggle to detect zero-day attacks and evolving threat patterns due to static training methods. [14]
- Gap 6: Explainability and Trust: Agentic AI systems often function as black-box models, reducing transparency in security-critical decisions. [13]

Open Questions:

- How to stabilize RL policy learning under adversarial pressure and certify behavior before autonomy in production? [6]
- How can Agentic AI systems enable reliable autonomous threat detection and response in real-world environments? [7], [13]
- How can Agentic AI integrate multimodal data sources (network logs, behavioral data, sensor data) to improve anomaly detection accuracy? [15]
- What architecture enables scalable multi-agent cybersecurity monitoring in distributed environments? [11]
- How can Agentic AI systems improve robustness against adversarial manipulation? [8]
- How can Agentic AI incorporate continuous learning to adapt to emerging cyber threats? [14]
- How can explainable AI techniques improve transparency and trust in autonomous cybersecurity systems?[13]

These gaps motivate the proposed research on developing scalable, autonomous, and adaptive Agentic AI-based cybersecurity threat detection systems.

1.14 Innovation and Conceptual Synthesis

The reference pattern used for finance is UEBA + CNN/LSTM/Transformer ensembles for early anomaly detection . For security implement agentic orchestration with limited access credentials, zero-trust policy, and SOAR.Sandboxed tool execution with protocol policy checks (MCP/A2A).Federated continual learning across providers.CSF 2.0-aligned controls (principle of least agency, immutable logs, dynamic revocation/quarantine). RL-assisted containment gated by safety metrics and human escalation. [1][2][3][5][6]

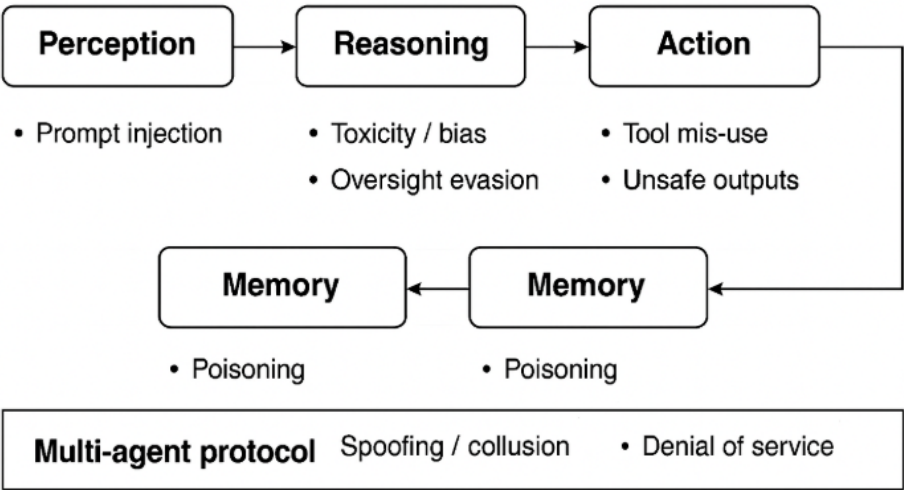


Figure 1. Agentic AI Threat Taxonomy (schematic)

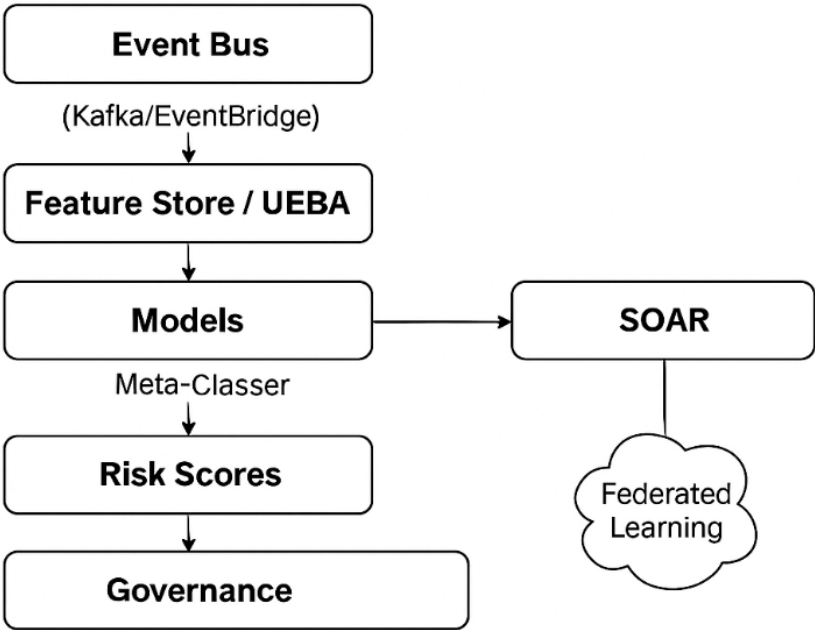


Figure 2. Event-Driven Agentic Security Architecture (Finance)

NIST CSF 2.0 → Agentic Controls

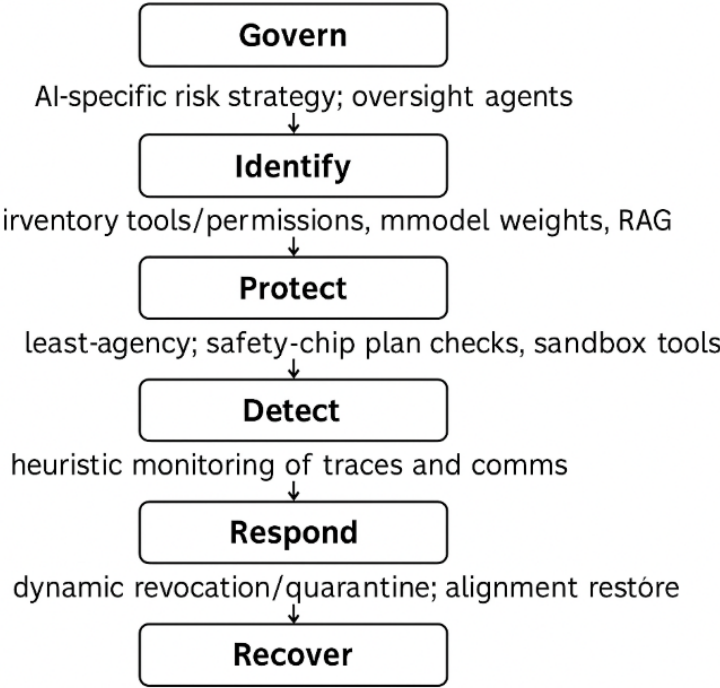


Figure 3. NIST CSF 2.0 Mapping for Agentic-AI Controls (schematic)

Adapted from the 2025 SLR that aligns agentic threats/controls with CSF 2.0 and OWASP model. [5]

Table 6. Mapping of NIST CSF 2.0 Functions to Agentic AI Security Controls

CSF 2.0	Agentic Focus	Representative Control	Refs
Govern	Strategy & oversight	Principle of Least Agency; oversight agents	[5]
Identify	Assets & risks	Inventory tools/permissions, model weights, RAG stores	[5]
Protect	Safe execution	Sandboxed tools; safety-chip plan checks; polymorphic prompts	[5]
Detect	Behavioral anomalies	Monitor tool sequences, reasoning traces, inter-agent comms	[5]
Respond	Containment	Dynamic revocation, quarantine, alignment restoration	[5]
Recover	Trusted state	Memory cleansing; rollback; blockchain-anchored audit	[5]

1.15 Visualizations and Knowledge Summaries

Figures 1–3 summarize core risks, a finance-centric architecture, and a governance mapping. Tables in Sections 1.8–1.10 consolidate techniques, benchmarks, and tooling for implementation and evaluation[1][2][3][5].

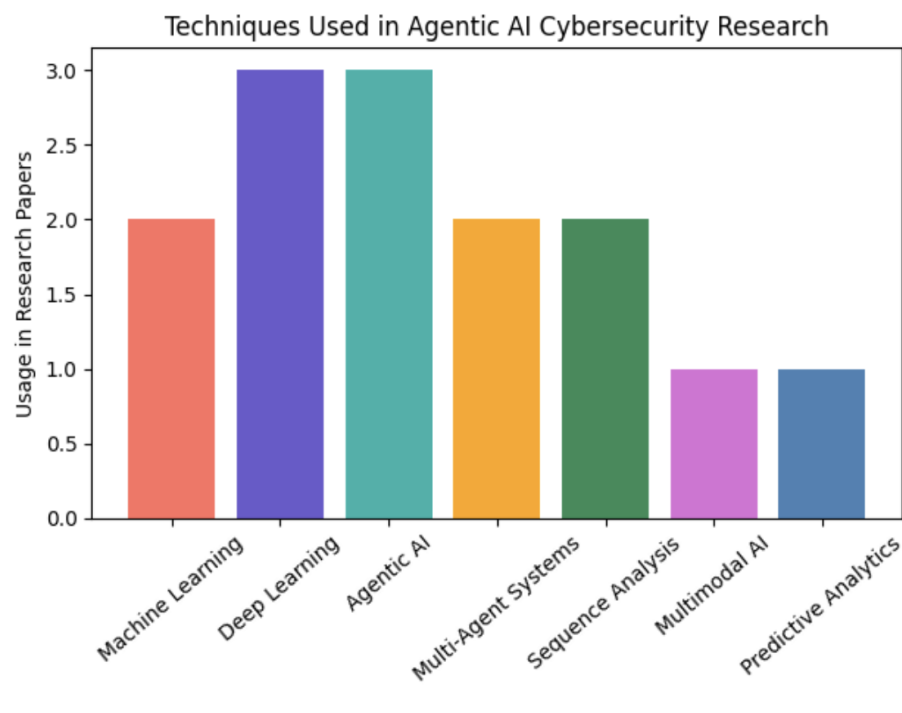


Figure 4. Techniques used in Agentic AI cybersecurity research

1.16 Academic Integrity and Citation Style

This chapter follows IEEE numeric citation style. Ideas are paraphrased with attribution to the sources; verbatim quotations are avoided or quoted with page/figure references where applicable [1]-[15].

1.17 Summary of the Chapter

Agentic AI can help us to improve detection and response through deep ensembles, UEBA, and orchestrated, policy-bound automation, especially in finance. Yet autonomy introduces threats at reasoning, action, memory, and multi-agent layers. CSF 2.0-aligned controls (least-agency, sandboxed tools, immutable logs, dynamic revocation) and process-aware evaluation are necessary complements. Reinforcement Learning adds responsiveness to the threat detection, but it needs stabilization and careful gating. [1][2][3][5][6]

1.18 Structure of the Report

Chapter 1: Introduction, Chapter 2: Requirements & Data, Chapter 3: Methods & Architecture, Chapter 4: Experiments & Evaluation, Chapter 5: Ethics, Governance, and Limitations, Chapter 6: Conclusions & Future Work.

2 Foundations

2.1 Introduction to the Foundations Chapter

The evolving landscape informs this research on cybersecurity in the context of advances in Machine Learning and Artificial Intelligence. Traditional security mechanisms that make use of Artificial Intelligence systems have found it difficult to adapt to the changing environment due to the presence of fixed rule sets and high dependency on human input [3], [4]. Thus, such security measures have proven to be insufficient in handling the ever-changing threat landscape. Considering the emergence of advanced forms of cyber-attacks using Artificial Intelligence systems, there arises an urgent requirement for self-regulating mechanisms.

Agentic Artificial Intelligence marks the evolution of this field, which is different from the concept of Artificial Intelligence. Agentic Artificial Intelligence is capable of providing long-term memory, planning, and tool-integration, which enables these tools to effectively operate in complex environments and respond to security threats in real time without the requirement for human intervention "in the loop" [1], [2]. Within this chapter, the algorithmic, theoretical, and architectural building blocks to design the entire security framework using Agentic Artificial Intelligence will be established.

2.2 Theoretical Underpinnings and Conceptual Framework

The present research employs a group of technical terms to describe the autonomous security framework. Agentic AI is described in this research as "a system in which cognition and execution can occur independently in uncertain environments" [1]. Moreover, this can be further supported by "User and Entity Behavior Analytics" (UEBA), which emphasizes "detecting anomalies by establishing baselines for human and machine entity behaviors" [3]. Finally, the implementation of defensive actions can be carried out through "SOAR" (Security Orchestration, Automation, and Response), which replaces human interventions through "high-speed workflows" [3].

From the perspective of data analysis, "anomaly" can be defined as "a significant deviation from a system's predicted behavior," which necessitates the use of "multimodal data" that combines different data streams such as "sensor data" and "network data" [15]. Moreover, the logic behind the agent's evolution can be supported by "reinforcement learning" (RL). In RL, "agents learn to transition between different states by executing actions according to policies that optimize reward signals from the environment" [6].

Agentic AI refers to intelligent systems capable of perception, reasoning, planning, and action execution in dynamic environments. Unlike traditional AI models, which are reactive, agentic systems are proactive and goal-oriented.

Anomaly Detection Theory Cyber threat detection is primarily based on identifying deviations from normal behavior.

Anomaly = $|X - \mu| > \theta$

Where:

X = observed behavior

μ = expected behavior

θ = anomaly threshold

This principle is extended using machine learning and deep learning models for complex environments.

Reinforcement Learning for Adaptive Defense

Agentic systems use reinforcement learning (RL) for decision-making:

Subsection 2.1. Introduction to the Foundations Chapter

Subsection 2.2. Theoretical Underpinnings and Conceptual Framework

Subsection 2.3. Terminology and Notational Conventions

Subsection 2.4. Overview of Problem-Solving Paradigms

Subsection 2.5. Algorithmic Foundations

Subsection 2.6. Classical Algorithms

Subsection 2.7. Modern Algorithmic Extensions

Subsection 2.8. Model Design Principles

Subsection 2.9. Model Architectures and System Components

Subsection 2.10. Input Processing Layer

Subsection 2.12. Core Processing Unit

Subsection 2.13. Output Generation Layer

Subsection 2.11. Data Preprocessing and Representation

Subsection 2.14. Feature Engineering and Selection Techniques

Subsection 2.15. Experimental Setup and Design Considerations

Subsection 2.16. Evaluation Metrics and Justification

Subsection 2.17. Domain-specific Constraints and Assumptions

Subsection 2.18. Interpretability and Explainability Foundations

Subsection 2.19. Implementation Toolchains (Conceptual View)

Subsection 2.20. Visualization Techniques and Diagrammatic Representation

Subsection 2.21. Ethical, Legal, and Societal Implications (ELSI)

Subsection 2.22. Challenges and Limitations in Theory or Practice

Subsection 3.12. Lessons from Iterative Development and Peer Feedback

Subsection 2.24. Responsible Use of Generative AI in Foundations Work

Subsection 2.25. Summary of the Chapter

$$\pi(\alpha \mid s) = P(\text{action} \mid \text{state})$$

Where:

s = system state

α = action

π = policy

RL enables agents to:

- Learn optimal responses
- Adapt to new attack strategies
- Minimize response time

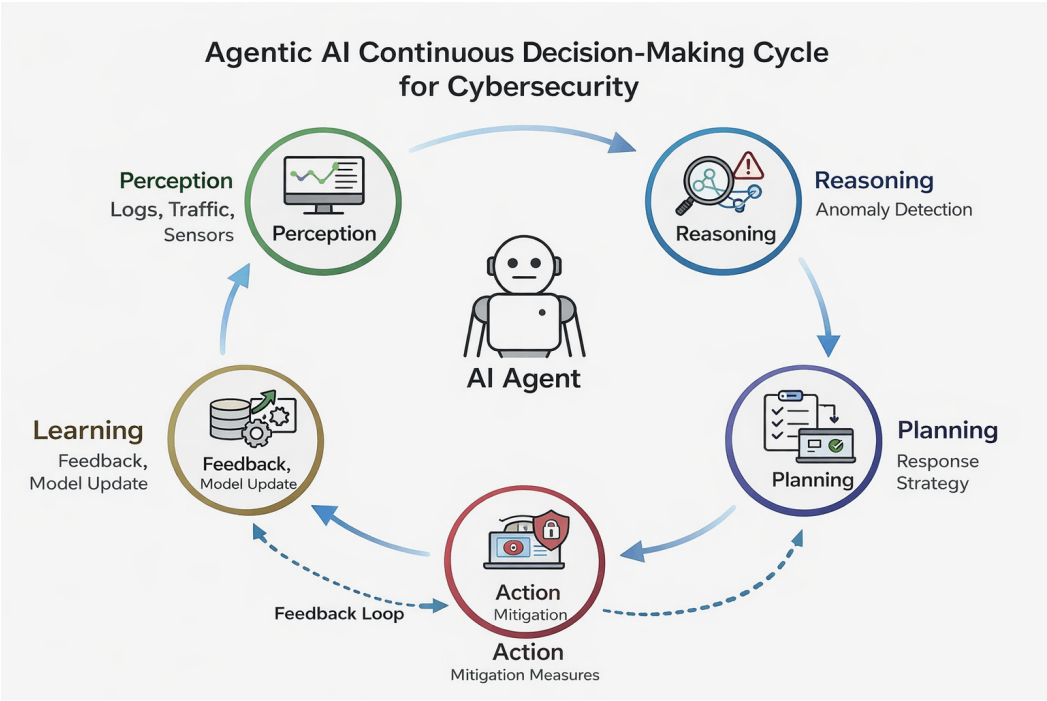


Figure 5. Decision Making Cycle

2.3 Terminology and Notational Conventions

This section we define technical terms, abbreviations, and symbols for consistency and clarity across the Chapter.

Table 7. Key Terms and Definitions

Term / Notation	Definition
Agentic AI	Autonomous AI system capable of reasoning, planning, and taking actions without continuous human intervention
Anomaly Detection	Process of identifying patterns in data that deviate from expected or normal behavior
UEBA (User and Entity Behavior Analytics)	Technique used to detect abnormal user or system behavior based on historical patterns
SOAR (Security Orchestration, Automation, and Response)	Platform that automates incident detection, investigation, and response workflows
Threat Intelligence	Information about potential or existing cyber threats used for proactive defense
Decision Engine	Core component that evaluates inputs and determines appropriate actions based on learned patterns
Feature Vector	Numerical representation of data used as input for machine learning models
False Positive (FP)	Incorrect identification of normal activity as a threat
False Negative (FN)	Failure to detect an actual threat
Precision	Ratio of correctly predicted positive observations to total predicted positives
Recall	Ratio of correctly predicted positive observations to all actual positives
F1-Score	Harmonic mean of precision and recall used to balance both metrics
PR-AUC	Area under the Precision-Recall curve, useful for imbalanced datasets
Reinforcement Learning (RL)	Learning paradigm where an agent learns by interacting with an environment using rewards and penalties
Multi-Agent System	System composed of multiple interacting intelligent agents working collaboratively or independently
SIEM	Security Information and Event Management

2.4 Overview of Problem-Solving Paradigms

Throughout history, the identification of cyber threats has followed the evolution of several technologically advanced stages. Early frameworks were designed using signature-based and rule-based logic. Although these frameworks ensured high transparency and processing speeds, they were unable to address the flexibility required to effectively counter the evolving exploits. Later, the application of machine learning technology ensured the adoption of a more analytical and data-centric approach. Even this technology was unable to overcome the challenges associated with the high dependency on manual feature selection and the inability to predict unknown zero-day attacks [9].

Another major shift in the evolution of the field came with the adoption of deep learning techniques. These techniques ensured the application of LSTMs, Transformers, and CNNs to automatically extract complex features from the huge streams of data. Today, the field is moving towards the application of agent-based AI, which is characterized by the application of entities. Unlike earlier approaches, this paradigm enables the application of constant surveillance and incident response, transforming the field from reactive to self-governing and proactive [7], [13].

Agentic AI represents an advanced paradigm where systems not only detect anomalies but also reason and take autonomous actions. This paradigm shifts the focus from reactive

defense to proactive and adaptive cybersecurity systems capable of operating in dynamic environments.

2.5 Algorithmic Foundations

The computational framework of this research includes sequence modeling, anomaly detection, and decision optimization. Classification and clustering are the primary methods for flagging behavioral anomalies in the system [3]. However, to cater to the time-constrained nature of cybersecurity data, LSTM and Transformer models are also being used. The former is particularly good at extracting temporal features, which are essential for detecting complex and non-linear forms of attacks [3].

Optimization of responses is also being carried out by implementing reinforcement learning. This allows autonomous agents to respond to changing adversarial behaviors by adapting their strategies [6]. Finally, data sovereignty is being addressed by implementing federated learning. This allows multiple organizations to train models and share intelligence without compromising data privacy and sovereignty [3].

2.6 Classical Algorithms

Typically, a traditional cybersecurity framework incorporates existing machine learning algorithms such as logistic regression, support vector machines, and decision trees. Such traditional models offer a high level of transparency and computational efficiency. Therefore, these traditional models serve as powerful tools for identifying known security risks [3].

However, despite the significant benefits of traditional machine learning models, these models have a fixed architecture that does not take into account the non-linear complexities of modern network data. The traditional models' use of human-based feature selection makes them inflexible tools for identifying new zero-day risks. Therefore, these traditional models are not sufficiently effective in identifying complex and evolving patterns of modern cyber threats [9].

2.7 Modern Algorithmic Extensions

Modern security strategies are a step forward from traditional strategies, incorporating deep learning and autonomous technologies. By employing deep ensembling techniques that combine CNN, LSTM, and Transformer models, these systems can achieve high accuracy. This is due to their capability to effectively handle both spatial and temporal features hidden in complex data flows [3].

The incorporation of agentic orchestration has taken a significant step toward advancing cybersecurity processes, introducing new dimensions of reasoning, planning, and tool-interaction capabilities. This takes a step closer to achieving autonomous decision-making processes in cybersecurity [1], [2]. To keep these systems at their peak, reinforcement learning techniques are employed to adapt defensive strategies based on environmental feedback [6]. The current state-of-the-art strategy in this domain is represented by hybrid models, which are complex frameworks incorporating anomaly-based and predictive analytics, along with agent-based reasoning capabilities [7], [13].

2.8 Model Design Principles

The design of the proposed framework is mainly guided by the need to ensure modularity, security, and scalability of the computations. This is because a modular system can allow the smooth integration of different functional layers of the system, including the detection

mechanism and the response mechanism. To ensure the scalability of the system and its ability to process large volumes of data, the proposed framework uses a distributed multi-agent system with the ability to perform parallel computations [11].

To ensure the integrity of the autonomous system, the proposed system uses a number of security protocols. This includes the use of sandboxing and the principle of least-agency to restrict the functionality of the system’s components [5]. To ensure the system’s ability to respond to the changing nature of the threat landscape, the system uses a number of continuous learning loops to refine the system’s ability to respond and react to the threat landscape [14].

2.9 Model Architectures and System Components

This section will provide a visual and textual breakdown of the system or model architecture used.

2.9.1 The proposed methodology comprises four core Agents:

Table 8. Agent Roles in the Proposed System

Agent	Role
Detection Agent	Utilizing hybrid machine learning models (CNN for network traffic analysis and Transformer-based models for log sequence analysis) to identify anomalies and extract indicators of compromise.
Investigation Agent	Employing natural language processing and knowledge graph reasoning to correlate threat intelligence from multiple sources and generate attack hypotheses.
Response Agent	Implementing policy-driven automated countermeasures such as endpoint isolation, firewall rule updates, and patch deployment.
Coordinator Agent	Managing inter-agent communication, priority assignment, and escalation workflows. The system leverages large language models (LLMs) integrated with cybersecurity tools through ReAct (Reasoning and Acting) frameworks and LangGraph for stateful agent orchestration.

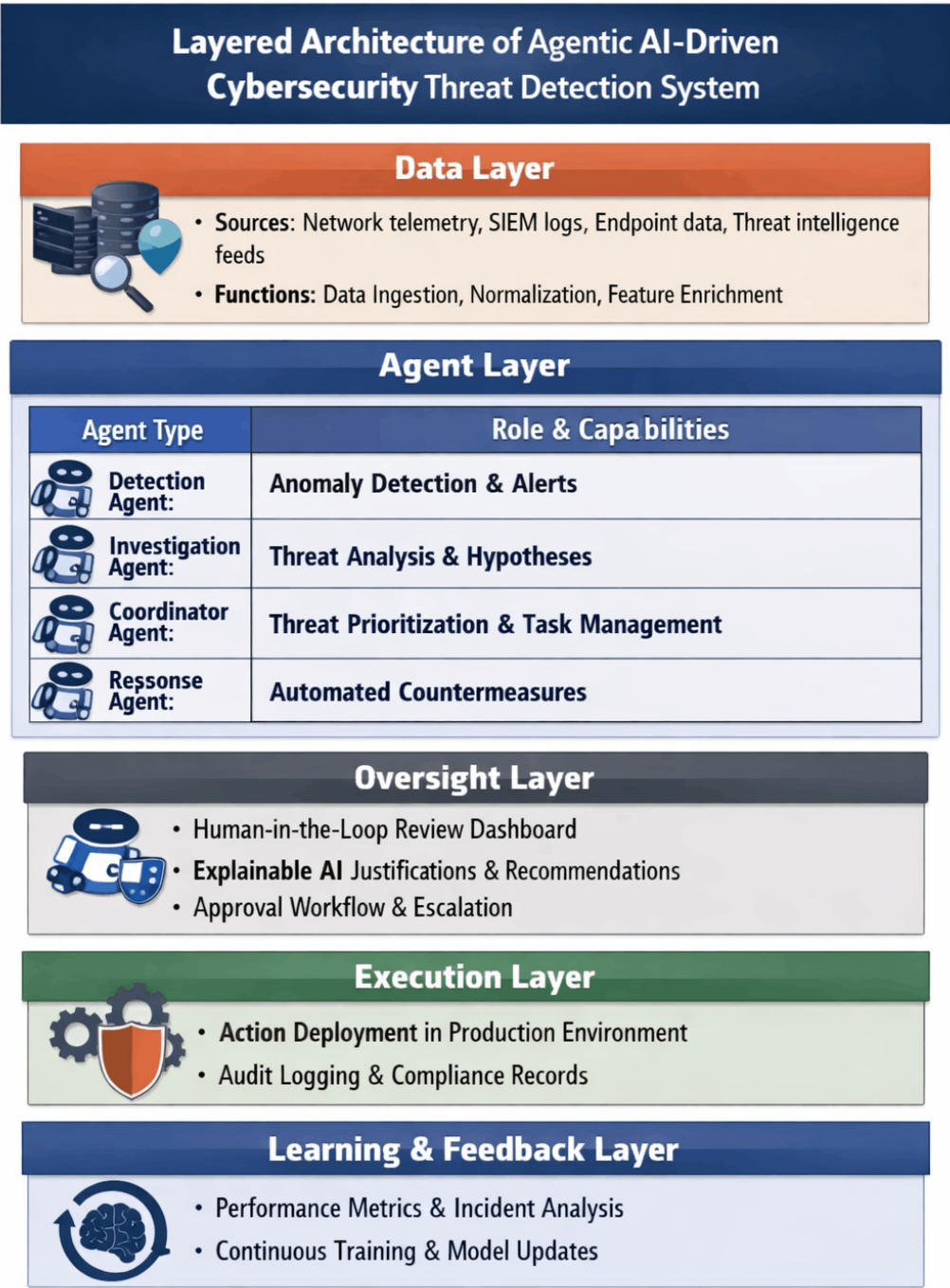


Figure 6. System Layers

2.9.2 High-Level Block Diagram for Proposed Methodology

System Workflow:

1. **Data Collection:** Aggregate security telemetry from network devices, SIEM systems, endpoints, and threat intelligence feeds
2. **Preprocessing:** Normalize heterogeneous data formats, extract relevant features, and enrich with contextual information

- 3. **Detection Phase:** Detection Agent analyzes data using hybrid ML models to identify anomalies and generate alerts with confidence scores
- 4. **Investigation Phase:** Investigation Agent correlates alerts with threat intelligence, performs root cause analysis, and generates attack hypotheses
- 5. **Coordination:** Coordinator Agent prioritizes threats, assigns tasks to specialized agents, and manages communication protocols
- 6. **Response Phase:** Response Agent executes automated countermeasures based on predefined policies and threat severity levels
- 7. **Oversight:** High-risk actions require human approval through oversight dashboard with explainable AI justifications
- 8. **Execution:** Approved actions are implemented in production environment with comprehensive logging for audit trails
- 9. **Feedback Loop:** System performance metrics and incident outcomes feedback into agent training for continuous improvement

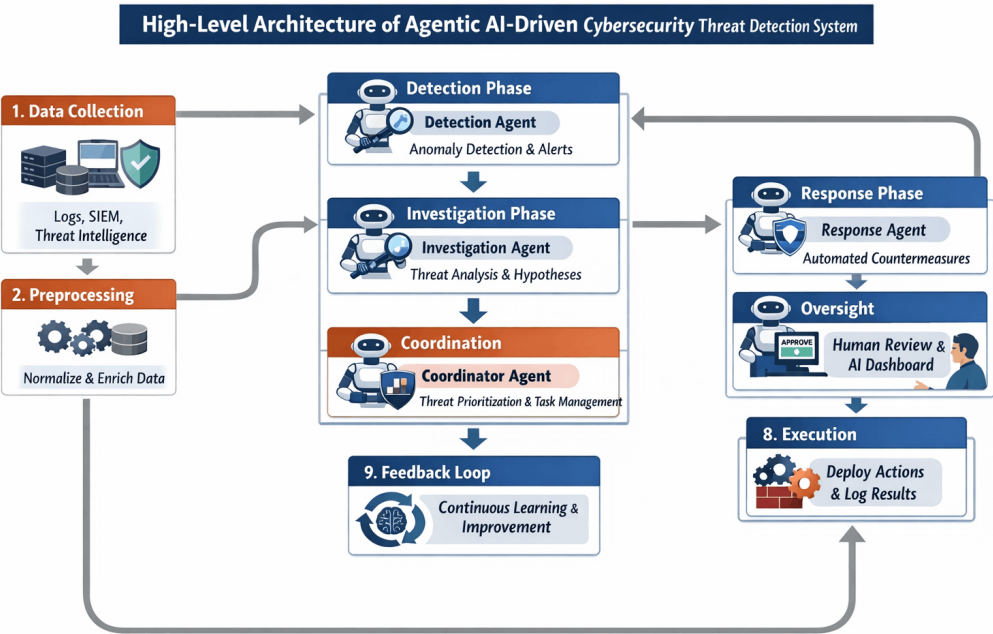


Figure 7. High Level Architecture

2.10 Input Processing Layer

The project has picked the Finance domain . The input processing layer will simulate the finance threats using the existing benchmark datasets as mentioned in the Literature review section 1.9.

This layer is considered to be the primary entry point for the system and is tasked with data collection and preprocessing from a variety of sources. It collects data streams from different sources, including network logs, system logs, and external threat intelligence feeds, and normalizes them. This layer aggregates security telemetry from network devices,

SIEM systems, endpoint, and threat intelligence feeds. Normalizing this data is considered a crucial step, as it allows unstructured data to be converted into structured data for computational analysis [15].

2.11 Data Preprocessing and Representation

The first part of this framework is concerned with the meticulous cleaning, normalization, and transformation of raw cybersecurity telemetry. This is an essential part of the framework, especially when considering the need to translate raw inputs into the system, such as raw logs. Through the use of automated log parsing and feature extraction, the framework is successful in translating raw logs into structured datasets that are necessary for machine learning analysis [15].

In order to further improve the analytical capabilities of the framework, complex data representation techniques are used. These techniques include sequence encoding and feature stores. These techniques are more than just mere data organization techniques. They are techniques that enable the framework to map out complex relationships and understand the essential time dependencies that are necessary in cybersecurity. Through this representation, the framework is able to identify even complex time-delay attacks [3].

2.12 Core Processing Unit

The brain of this framework, so to speak, is the core processing unit, a central core through which anomaly detection models, agentic reasoning models, and orchestration models converge. This framework, therefore, uses deep learning models to detect anomalies from the norm, while the agentic models assess the environment to determine what course of action would be most appropriate. To make this framework scalable and able to perform distributed oversight through a growing network, a multi-agent architecture has been incorporated, allowing for both scalability and parallel processing ability [11], [13].

2.13 Output Generation Layer

The last layer, namely the Output Generation Layer, has the responsibility of converting the result obtained from the core unit into a form that can be implemented, such as risk evaluation, security alerts, and even defensive actions. This layer has a direct interface with Security Orchestration, Automation, and Response tools, ensuring that actions are implemented in a prompt manner without requiring human intervention. However, the system does not stop at just implementing actions; rather, it also seeks to ensure transparency and accountability through a complete reasoning trail and decision logs, ensuring that all actions implemented by the system are "explainable" and can be traced by security teams to understand the logic behind a particular mitigation strategy [3].

2.14 Feature Engineering and Selection Techniques

However, the process of feature engineering is the core of converting the raw telemetry into intelligent output. This process entails the meticulous extraction of meaningful patterns, which can range from high-level behavioral characteristics to detailed statistical metrics and intricate temporal patterns [14]. To ensure the system remains computationally lean without compromising efficiency, complex feature selection methods are implemented. This process filters out the noise and the most impactful data points, thus directly increasing the efficiency of the model.

Although the process of manual engineering is a structured approach, the inclusion of deep learning models has proven to be a significant leap in this regard. This is because

deep learning models can automatically extract features, which significantly reduces the dependency on traditional parameters. This process can automatically detect intricate and non-linear relationships between the data points, thus significantly increasing the accuracy of the system and its ability to detect novel threats [3].

2.15 Experimental Setup and Design Considerations

The research architecture of this study is based on a robust validation approach, which begins with the systematic splitting of the available datasets into separate training sets, validation sets, and testing sets. To guarantee the robustness and reliability of the developed models, the approach uses cross-validation methods. This is important in reducing the risk of overfitting while maintaining the consistency of the system's prediction capabilities with different and unseen data profiles [14].

Besides data splitting, there are some essential design aspects incorporated to ensure data integrity. These aspects include tackling data imbalance issues, which is a major problem for cybersecurity datasets. It is followed by data quality and experiment reproducibility. In order to bridge the gap between static analysis and practical application, this study utilizes simulation environments. These environments allow for the safe exploration of autonomous agent behaviors, giving a chance to test how agents reason and respond to complex threats before practical application [1], [2].

2.16 Evaluation Metrics and Justification

In order to fully understand the efficacy of the proposed system, a framework has been developed and categorized into three different categories, namely detection, safety, and reliability. The detection metrics are considered to be the main criteria for measuring the efficacy of the system in identifying anomalies effectively. This includes standard metrics such as precision, recall, F1 score, and Area Under Precision Recall Curve, which enable a fine-grained evaluation of the accuracy of the system in diagnosing anomalies effectively [3].

In addition to detection, safety metrics are also incorporated in the framework to assess how effectively the system performs under pressure. Metrics such as policy compliance, risk ratio, and attack success rate are critical in measuring how effectively the system performs under pressure and how effectively autonomous actions are carried out within authorized boundaries [2], [5]. To make the framework more viable, reliability metrics are incorporated to assess how effectively the system performs under repeated trials, ensuring that its efficacy does not reduce over time or under different environmental conditions [1].

2.17 Domain-specific Constraints and Assumptions

The proposed system has been particularly designed with the aim of deployment in the critical domain of financial cybersecurity. The framework has to operate in a challenging environment where real-time data processing, regulatory compliance, and data privacy are critical non-negotiables in financial cybersecurity, as discussed in [3] and [4]. The proper functioning of the proposed model relies upon a series of underlying principles, including the presence of logging mechanisms, structured data, and adequate computing capabilities.

Additionally, the rationale behind the entire framework is based on the understanding that security threats are always characterized by certain behavioral patterns deviating from the norm. This enables the system to detect these patterns and possibly alert the system to potential security breaches or fraud attempts. To guarantee the effectiveness of the system in the face of adaptive threats, the entire architecture is predicated on the possibility of

applying adaptive learning techniques to improve the accuracy of the system's detection capabilities and evolution [14].

2.18 Interpretability and Explainability Foundations

In the critical domain of cybersecurity, explainability is no longer just an attribute, but rather a fundamental requirement in order to gain human trust in automated decision-making. One major challenge facing contemporary cybersecurity has been that many state-of-the-art deep learning architectures tend to be "black boxes," where the underlying reasoning behind a particular threat detection mechanism remains obscure and hard to interpret by security analysts [13]. In order to address this challenge, agentic systems have been designed with a layered approach to transparency. By providing detailed reasoning traces, extensive action logs, and strong auditing capabilities, agentic systems offer a "paper trail" that explains every autonomous decision taken by the system.

In order to further enhance human-in-the-loop decision-making in critical domains, the framework has been extended with a suite of Explainable AI (XAI) techniques. These techniques will allow security analysts to interpret complex neural network patterns, thus demystifying the underlying reasoning behind a particular threat detection mechanism. By providing human analysts with detailed reasoning traces, security analysts will be able to verify whether autonomous decision-making has been justifiable and transparent, rather than simply expedient.

2.19 Implementation Toolchains (Conceptual View)

The practical implementation of the proposed system is grounded in an already well-developed Python-based system, utilizing well-known and well-supported deep learning frameworks such as PyTorch and TensorFlow. However, to take the system beyond the realm of static detection, the system is designed to integrate well-known and well-supported multi-agent frameworks, designed to enable true agentic capabilities. The frameworks are then augmented by the implementation of Large Language Models (LLMs), acting as the cognitive core, responsible for the high-level reasoning and decision-making necessary to effectively navigate the complex world of security [15].

To handle the high levels of computational power required by modern cybersecurity, the system is designed to take advantage of cloud computing and distributed system technologies. The infrastructure is designed to ensure the horizontal scalability necessary to handle the high levels of data throughput and analysis without sacrificing system performance [9]. Supporting the infrastructure is an array of automated data preprocessing and monitoring tools, designed to handle the end-to-end lifecycle of data management. By streamlining everything from the initial data ingestion to the end-state performance monitoring, the implementation is designed to ensure that the agentic logic is supported by clean, relevant, and timely data.

2.20 Visualization Techniques and Diagrammatic Representation

However, to effectively translate the complex security data into meaningful and actionable knowledge, the framework leverages a wide variety of visualization methods. This starts with the high-level system architectures, threat taxonomies, and governance mappings, which give a broad understanding of the system's structure and the relationships between the various components of the security system. This provides a better understanding of the relationships between the different components of the system and the nature of the risks they are exposed to.

However, to effectively operate the system in real-time, the system uses a number of graphical representations of the system and the various components of the system. This is particularly important because it is specifically designed to highlight anomalies and system performance in a manner that is immediately intuitive and understandable. This provides a significant enhancement of the system’s awareness and enables rapid and effective decision-making in the cybersecurity system [3]. Whether it is the detection of a sudden spike in irregular traffic patterns or auditing the effectiveness of the autonomous system’s response, these visual representations are a critical link between the raw data and the oversight process.

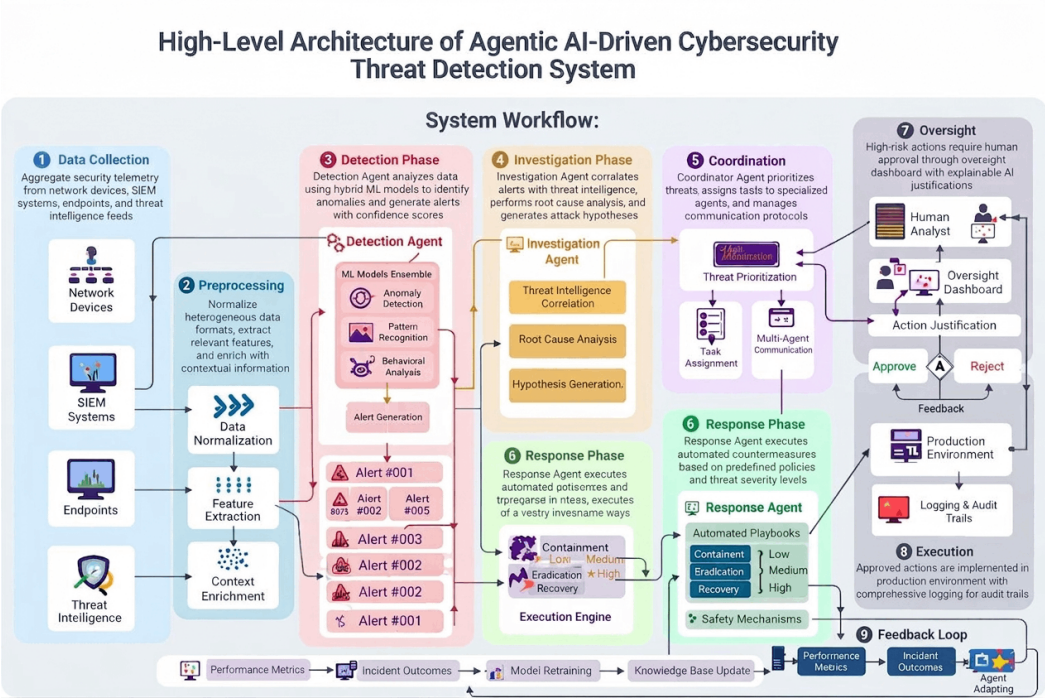


Figure 8. System WorkFlow

2.21 Ethical, Legal, and Societal Implications (ELSI)

The incorporation of Agentic AI in the domain of cybersecurity has given rise to a complex gamut of ethical and legal dilemmas, primarily related to data privacy, algorithmic bias, and the underlying principle of accountability itself [8]. This is due to the fact that, owing to its autonomous nature, Agentic AI cannot operate in a vacuum and hence has to be aligned with existing regulatory frameworks, ensuring not only its efficacy but also its safety and socially acceptable nature.

In order to avert any potential risks associated with Agentic AI, a high degree of governance frameworks has to be incorporated, ensuring that there is no scope for its potential misuse and a high degree of transparency, especially in critical industries such as finance [5]. This can be ensured through a proper framework of protocols, ensuring a fine line between the rapid response nature of Agentic AI and its underlying human-defined nature.

2.22 Challenges and Limitations in Theory or Practice

The move towards autonomous security is also faced with a number of challenges, such as those concerning datasets, scalability, and generalization. The major problem is that

a number of datasets, especially those used for cybersecurity, are no longer valid, as they lack the modern diversity required to effectively train AI models for modern, real-world scenarios. This "data aging" is a major problem, as it limits the potential for AI systems to effectively respond to modern, high-sophistication cyber-attacks [14].

In addition to these challenges, the use of agentic systems is also faced with a number of operational risks, such as the potential for adversarial manipulation, where an attacker may use the AI logic to their own advantage, and the potential for "unsafe" actions, especially in a crisis situation. Finally, the complexity that arises when the number of autonomous systems increases is a major problem, often resulting in synchronization conflicts and poor resource allocation [1], [2], [5].

2.23 Lessons from Iterative Development and Peer Feedback

The path to an effective high-performance cybersecurity system is described in terms of an iterative development process and an evaluation cycle. While this is not meant to be taken literally, it is more of a figure of speech to describe the continuous improvement process based on the outputs of rigorous experiments and standardized benchmarking. Through the analysis of the performance outputs, specific methodological limitations are also identifiable, thus allowing for improvements to be made to the efficacy of the cybersecurity system [1].

In addition to the internal improvement process of the cybersecurity system, the influence of the scientific community on the development of cybersecurity systems is also significant. Peer-reviewed research is essential to provide the necessary validation for novel approaches to cybersecurity. Through the scrutiny of novel methodologies, cybersecurity systems are developed on the foundation of reliability and empirical evidence. This shared knowledge does not just contribute to the improvement of individual cybersecurity systems but also to the improvement of the entire industry to develop more robust, transparent, and trustworthy AI-driven cybersecurity solutions.

2.24 Responsible Use of Generative AI in Foundations Work

The introduction of generative AI in the cybersecurity world also comes with a tremendous responsibility to adhere to ethical standards and academic integrity. To this end, and in order to fully comply with all relevant regulations, there has to be a commitment to transparency and proper citation of all foundational work in its implementation and use [8]. This serves as a foundation for a professional and credible approach to security research, especially in a world in which automated content generation can so easily be used to mask true origins and intent.

Other than these ethical considerations, however, the philosophy behind the operation and use of this system revolves around the concept of human augmentation. This means, rather than seeing generative AI as a replacement for human intervention, we believe that its most effective use comes in the form of a sophisticated aid to human decision-making, especially in a security world in which nuanced human judgment, intuition, and accountability are absolutely critical and cannot be replaced by a machine, no matter how advanced.

2.25 Summary of the Chapter

As can be seen, this current chapter has established the foundational theoretical, algorithmic, and architectural underpinnings necessary to enable an emerging Agentic AI-based cybersecurity paradigm. One of the overarching themes of this current work has been the

need to transition away from legacy rule-based systems that are static and circumventable, towards more autonomous and adaptive models. These agentic models are uniquely capable of real-time threat identification and response, thus affording a more dynamic defense against the ever-increasing threat landscape of digital risks [7], [13].

However, while the advantages offered in terms of automation and flexibility are obvious, there exist certain complexities associated with the deployment of Agentic AI. The move towards autonomy has given rise to a series of systemic challenges, especially in relation to operational safety, regulatory governance, and decision explanation. The only way to overcome these complexities is through a multi-dimensional approach that incorporates structured technical frameworks, along with evaluation procedures and ethical considerations. The only way to effectively balance these high-speed automated capabilities with human-defined guardrails is to ensure the development of reliable next-generation cybersecurity systems, which would be required in the future [1], [2], [5].

3 Interim Report

3.1 Chapter Overview

Briefly introduce the purpose of the interim report, scope of progress covered, and its role in shaping the remainder of the project.

3.2 Dataset Acquisition and Preprocessing

3.2.1 Dataset Description

Detail the datasets used, including source, structure, size, and features.

3.2.2 Data Quality Checks

Highlight steps taken to handle missing values, duplicates, or inconsistencies.

3.2.3 Exploratory Data Analysis (EDA)

Discuss trends, distributions, and insights from the dataset using charts and statistics.

3.2.4 Preprocessing Pipelines

Describe encoding, normalization, augmentation, or transformation techniques applied.

3.3 Feature Engineering

3.3.1 Feature Identification Techniques

Explain how features were selected or extracted from raw data.

3.3.2 Dimensionality Reduction (if applicable)

Discuss use of PCA, LDA, or other techniques to reduce feature space.

3.3.3 Feature Impact Observations

Summarize which features appear influential in preliminary testing.

3.4 Technical Implementation

3.4.1 Tools and Technology Stack

List the platforms, languages, libraries, and IDEs used so far.

3.4.2 System Setup and Environment Configuration

Explain the computing environment, OS, dependencies, GPU usage, version control.

3.4.3 Modular Codebase Description

Document implemented modules, APIs, or pipelines (with screenshots if applicable).

3.5 Baseline Model Development

3.5.1 Baseline Algorithm Description

Describe the first version of your model or algorithm.

3.5.2 Training Setup and Parameters

List hyperparameters, epochs, batch sizes, optimizers, etc., used in initial experiments.

3.5.3 Early Performance Metrics

Present results like accuracy, precision, F1-score, latency, etc., from the baseline runs.

3.5.4 Output Visualization and Interpretation

Include confusion matrices, ROC curves, or output plots to explain behavior.

3.6 Preliminary Analysis and Observations

3.6.1 Result Analysis and Discussion

Interpret model behavior and accuracy trends; identify possible sources of error.

3.6.2 Error Profiling and Bias Detection

Mention misclassifications, overfitting signs, or bias in predictions.

3.6.3 Lessons Learned

Discuss initial surprises, technical challenges, or confirmations of earlier hypotheses.

3.7 Originality and Innovation in Approach

3.7.1 Creative Framing of the Problem

Highlight unique perspectives or constraints applied in your approach.

3.7.2 Experimental Innovations

Propose alternative techniques, architectures, or data-handling strategies explored.

3.8 Risk Assessment and Project Management

3.8.1 Progress Tracker vs. Original Timeline

Show Gantt chart or milestone chart with updates.

3.8.2 Bottlenecks and Delays

Mention technical or resource-related delays and how they are being addressed.

3.8.3 Risk Mitigation Strategies

Identify potential risks in computation, ethics, or data and your contingency plans.

3.9 Experimental Reproducibility

3.9.1 Data Splits and Control Measures

Clearly state train/validation/test splits and their reproducibility strategy.

3.9.2 Random Seed Control

Mention if random seeds, initialization settings, or sampling techniques are made repeatable.

3.10 Scalability and Efficiency Considerations

3.10.1 Memory, Time, and Resource Profiling

Mention compute time, CPU/GPU usage, memory consumption.

3.10.2 Scalability Constraints

Highlight algorithmic limitations for larger datasets or real-time deployment.

3.11 Ethical Considerations and Responsible AI Use

3.11.1 Generative AI Usage Disclosure

Disclose if tools like ChatGPT, Copilot, Bard, etc., were used in writing, coding, or planning.

3.11.2 Bias and Fairness in Dataset

Comment on demographic fairness, sampling bias, or ethical concerns in your dataset or model.

3.12 Feedback Incorporation and Continuous Improvement

3.12.1 Peer and Mentor Feedback Summary

List key feedback points and how they influenced current progress.

3.12.2 Reflection and Learning Journey

Describe skills acquired, mindset changes, and personal development as a researcher.

3.13 Chapter Summary and Next Steps

Summarize interim progress, major takeaways, and refinements planned for the final phase of the project.

Optional Appendices Visuals: Confusion matrices, EDA plots, timelines

Tables: Detailed metric breakdowns, hyperparameter logs

Reproducibility: Scripts, requirements.txt, model cards

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